KnowFeat: Knowledge-Guided Feature Engineering with LLM Agents and Multi-Stage Verification

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***Abstract*—Automated feature engineering (AutoFE) is critical for improving the performance of machine learning models on tabular data. Recent approaches leverage large language models (LLMs) to generate features, but they rely solely on the LLM's internal knowledge and lack structured domain expertise, rigorous verification, and traceability. In this paper, we propose KnowFeat, a knowledge-guided feature engineering framework that integrates hierarchical domain knowledge into an LLM-based agent to generate high-quality, interpretable features. KnowFeat introduces three key innovations: (1) a *five-level knowledge hierarchy* ranging from schema-only metadata to risk indicators, detection rules, expert opinions, and court document evidence; (2) a *three-stage verification pipeline* (L1: execution, L2: statistical quality, L3: model effectiveness) that ensures only genuinely useful features are accepted; and (3) a *SHAP-guided search strategy* with failure memory and forward selection to efficiently explore the feature space. We evaluate KnowFeat on seven public datasets and a real-world anti-money laundering (AML) dataset with 200K transactions. On public benchmarks, KnowFeat achieves the best AUC on 4 out of 7 datasets and the best average rank (1.4) among seven methods including OpenFE, AutoFeat, CAAFE, FeatLLM, and LLM-FE, while generating 3–10× fewer features than FeatLLM. On the AML dataset, 18 knowledge-guided features improve Recall by +2.1pp and F1 by +1.5pp over a strong 93-feature baseline, outperforming methods that add up to 1,031 features. Ablation studies confirm that domain knowledge is essential, and every generated feature is traceable to specific knowledge assets.**

***Index Terms*—Feature Engineering, Large Language Models, Domain Knowledge, Anti-Money Laundering, Tabular Data**

I. Introduction

Feature engineering is widely recognized as one of the most impactful steps in building machine learning models for tabular data [1]. Well-crafted features can dramatically improve model accuracy, reduce overfitting, and provide interpretable signals for downstream tasks. However, manual feature engineering is labor-intensive, requires deep domain expertise, and does not scale across diverse applications.

To address this challenge, automated feature engineering (AutoFE) methods have been developed. Traditional AutoFE approaches such as OpenFE [2] and AutoFeat [3] apply systematic transformations—mathematical operations, aggregations, and feature crosses—to existing columns, then select the most predictive ones. While effective, these methods are purely data-driven: they explore a predefined transformation space without any understanding of what the features *mean* or why certain transformations might be useful for a given domain.

The emergence of large language models (LLMs) has opened a new paradigm for feature engineering. Methods such as CAAFE [4] and FeatLLM [5] use LLMs to generate feature transformation code based on dataset descriptions. These approaches leverage the LLM's broad world knowledge to propose semantically meaningful features—for example, generating a "debt-to-income ratio" for a credit dataset without being explicitly told to do so. However, current LLM-based methods suffer from three fundamental limitations:

(1) No structured domain knowledge. Existing methods rely entirely on the LLM's parametric knowledge, which is general-purpose and may be outdated, incomplete, or irrelevant to specialized domains. In high-stakes applications such as anti-money laundering (AML), the relevant knowledge—regulatory indicators, detection rules, expert judgment, and evidence from legal proceedings—exists in structured external sources that LLMs cannot access through their training data alone.

(2) Weak or absent verification. CAAFE evaluates candidates with a simple logistic regression inside the generation loop, while FeatLLM performs no model-based validation at all. Neither method checks whether generated code executes correctly, whether features exhibit meaningful statistical properties, or whether they genuinely improve the target model's performance. This leads to the acceptance of noisy, redundant, or even harmful features.

(3) No traceability. In regulated industries such as finance, healthcare, and legal compliance, model features must be auditable. Decision-makers need to understand *why* a feature was created and *what evidence* supports its relevance. Existing methods output code snippets without any provenance information, making them unsuitable for deployment in environments that require explainability and accountability.

In this paper, we propose **KnowFeat** (**Know**ledge-guided **Feat**ure engineering), a framework that addresses all three limitations. KnowFeat has three core components:

- **Hierarchical Knowledge Injection.** We organize domain knowledge into five levels of increasing specificity: (i) schema-only metadata, (ii) risk indicators, (iii) detection rules, (iv) expert opinions, and (v) evidence extracted from court documents. This hierarchy is *ablatable*: we can systematically study the contribution of each knowledge level. The LLM agent receives the relevant knowledge as structured context and is instructed to ground every generated feature in specific knowledge assets.
- **Three-Stage Verification Pipeline.** Each candidate feature passes through three verification stages. *L1 (Execution)*: the generated code is executed in a sandbox to

ensure correctness. *L2 (Statistical Quality)*: the feature is checked for information value (IV), Kolmogorov–Smirnov (KS) statistic, redundancy with existing features, and missing rate. An automatically calibrated noise threshold (from 100 random features at the 95th percentile) prevents low-quality features from proceeding. *L3 (Model Effectiveness)*: the feature is evaluated with XGBoost on the actual target metric across 5 random seeds, ensuring that only features with a genuine positive contribution are accepted.

- **Intelligent Search Strategy.** We decompose the feature space into semantic dimensions and allocate a per-round budget to each. After each round, SHAP analysis identifies which dimensions have contributed least, and the budget is reallocated inversely—directing the agent to explore underrepresented areas. A failure memory prevents the regeneration of previously rejected features, while a forward selection mechanism discovers feature combinations that are individually weak but collectively strong.

We evaluate KnowFeat on seven public tabular datasets from OpenML and a real-world AML dataset containing 200,000 digital currency transactions with 93 baseline features. We compare against six baselines spanning traditional AutoFE (OpenFE, AutoFeat), LLM-based (CAAFE, FeatLLM), and evolutionary LLM (LLM-FE) approaches. Key findings include:

- On public datasets, KnowFeat achieves the highest AUC on 4 out of 7 datasets and the best average rank (1.4) among all seven methods, while generating 3–10$\times$ fewer features than FeatLLM.
- On the AML dataset, 18 knowledge-guided features improve Recall by +2.1pp and F1 by +1.5pp over a strong 93-feature baseline, substantially outperforming all baselines including LLM-FE (+1.4pp F1).
- Ablation experiments confirm that domain knowledge is essential: the schema-only variant generates zero accepted features on one dataset, and the full knowledge hierarchy outperforms schema-only on 5 of 6 remaining datasets.
- Every generated feature includes a provenance card linking it to specific knowledge assets, providing full auditability.

The remainder of this paper is organized as follows. Section II reviews related work. Section III presents the KnowFeat framework. Section IV describes our experimental setup and results. Section V illustrates KnowFeat's knowledge-grounded features through case studies. Section VI concludes with limitations and future directions.

II. Related Work

A. Traditional Automated Feature Engineering

Automated feature engineering aims to reduce the manual effort of creating predictive features from raw data. Early approaches focus on exhaustive transformation search: Deep Feature Synthesis [6] recursively applies aggregation and transformation primitives along relational paths, while ExploreKit [7] enumerates unary, binary, and higher-order operators over existing features and uses meta-learning to rank candidates. AutoFeat [3] constructs features through polynomial and interaction terms, then applies L1 regularization for selection. More recently, OpenFE [2] frames feature generation as an operator-operand search problem, using a two-stage expansion-and-selection pipeline with LightGBM-based feature importance to achieve state-of-the-art performance on tabular benchmarks.

These methods share a common limitation: they operate in a *syntactic* transformation space, applying predefined mathematical operators without any semantic understanding of the data domain. For example, computing `age × income` is treated identically to `age × zip_code`, even though only the former has financial meaning. This leads to a combinatorial explosion of candidates, most of which are semantically meaningless, and an inability to generate domain-specific features that require conceptual understanding (e.g., "transaction velocity" or "dormant account reactivation").

B. LLM-Based Feature Engineering

The application of LLMs to feature engineering represents a paradigm shift from syntactic to semantic feature generation. CAAFE [4] iteratively prompts an LLM with dataset descriptions and evaluation feedback to generate Python code for new features. Each generated feature is evaluated using a logistic regression model, and the feedback is used to guide subsequent iterations. FeatLLM [5] takes a two-pass approach: first extracting natural language rules from the LLM, then converting these rules into binary indicator features.

More recently, LLM-FE [8] frames feature engineering as an evolutionary optimization problem, using LLMs to iteratively generate and refine features through structured prompts with performance feedback.

While these methods demonstrate the potential of LLMs for feature engineering, they share significant shortcomings. First, they rely exclusively on the LLM's parametric knowledge, which is a frozen snapshot of general web text that may lack domain-specific expertise. Second, their verification mechanisms are rudimentary: CAAFE uses a weak proxy model (logistic regression), FeatLLM performs no model-based validation, and LLM-FE relies on downstream model accuracy without statistical quality checks. Third, the generated features come with no provenance information—there is no record of *why* a feature was created or what knowledge supports it.

Concurrent work has explored LLMs for related data science tasks. AIDE [9] uses LLM agents to automate the entire data science pipeline, and DS-Agent [10] adopts a similar end-to-end approach with case-based reasoning. These methods do not address knowledge-grounded feature generation and provenance tracking.

C. Domain Knowledge Integration in Machine Learning

The integration of domain knowledge into machine learning has a long history. Knowledge-infused learning [11] incorporates external knowledge graphs into neural architectures. In the feature engineering context, domain experts traditionally

TABLE I
COMPARISON OF AUTOMATED FEATURE ENGINEERING METHODS.

Property	OpenFE	Auto. FE	CAAFE	Feat. Wiz	LLM-FE	Ours
Semantic underst.			✓	✓	✓	✓
External knowl.						✓
Exec. verific.	–	–				✓
Statistical filt.	✓	✓				✓
Model validation			Weak		✓	✓
Provenance track.						✓
Adaptive search					✓	✓

encode their knowledge as hand-crafted features—a process that is effective but does not scale. Transfer learning approaches [12] implicitly transfer knowledge through pretrained representations, but these are primarily designed for unstructured data (text, images) rather than tabular features.

In the financial domain specifically, AML feature engineering relies on expert-defined indicators such as transaction velocity, structuring patterns, and network centrality measures [13]. Regulatory bodies publish typologies and red-flag indicators (e.g., FATF guidelines), and financial institutions maintain internal detection rules derived from compliance experience and legal proceedings. This rich, structured knowledge has not been systematically leveraged by automated feature engineering methods.

D. LLM Agents with Tool Use

The ReAct framework [14] demonstrated that LLMs can interleave reasoning and action through tool use, enabling more reliable problem-solving than pure generation. Subsequent work has applied tool-augmented LLM agents to code generation [15], data analysis [16], and scientific discovery [17]. KnowFeat builds on this paradigm by equipping the LLM agent with specialized tools for data exploration, feature generation with provenance, and multi-stage verification—creating a closed-loop system where the agent can iteratively refine its feature proposals based on concrete feedback.

E. Positioning of KnowFeat

Table I summarizes the key differences between KnowFeat and existing approaches. KnowFeat is the first method to combine structured external domain knowledge, multi-stage verification (L1–L3), and full feature provenance tracking within an LLM-based feature engineering framework.

III. METHODOLOGY

This section presents the KnowFeat framework in detail. We first provide an overview of the pipeline (Section III-A), then describe each component: hierarchical knowledge injection (Section III-B), the ReAct agent with tool use (Section III-C), the three-stage verification pipeline (Section III-D), and the intelligent search strategy (Section III-E).

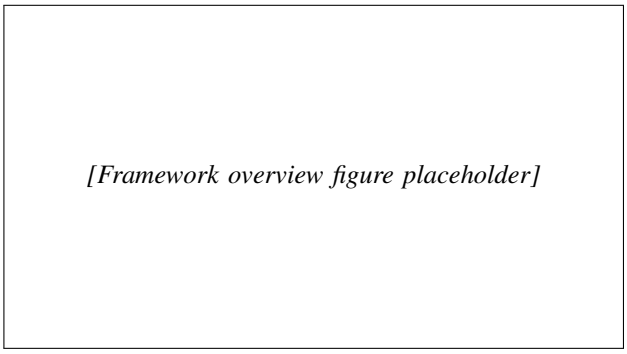


Fig. 1. Overview of the KnowFeat framework. The LLM agent generates features guided by hierarchical domain knowledge. Each candidate is verified through a three-stage pipeline (L1–L3). SHAP feedback dynamically reallocates the exploration budget across semantic dimensions.

A. Framework Overview

KnowFeat is a multi-round, knowledge-guided feature engineering framework built on an LLM agent. Given a tabular dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$ with baseline features $\mathbf{x}_i \in \mathbb{R}^d$ and target label y_i, along with a domain knowledge base $\mathcal{K}$, KnowFeat generates a set of new features $\mathcal{F}^* = \{f_1, f_2, \ldots, f_k\}$ such that the augmented dataset $\mathcal{D}' = \{(\mathbf{x}_i \| f_1(\mathbf{x}_i) \| \cdots \| f_k(\mathbf{x}_i),\ y_i)\}$ achieves improved predictive performance on a target metric (e.g., AUC or Recall).

The overall pipeline operates as follows (see Figure 1 for an illustration):

1) **Initialization.** Load the dataset, compute baseline model performance, and calibrate statistical thresholds via noise injection (Section III-D).
2) **Knowledge Assembly.** Organize domain knowledge $\mathcal{K}$ into a five-level hierarchy and inject the selected level into the agent's context (Section III-B).
3) **Multi-Round Generation.** For each round $r = 1, \ldots, R_{\max}$:
   a) Allocate per-dimension budgets based on SHAP feedback from the previous round.
   b) For each dimension, the LLM agent explores the data, proposes candidate features grounded in knowledge assets, and submits them for verification.
   c) Each candidate passes through the L1→L2→L3 pipeline. Accepted features are added to the baseline; rejected features enter the failure memory.
   d) Forward selection discovers feature combinations that are individually weak but collectively effective.
4) **Convergence.** The process terminates when no new features are accepted for C consecutive rounds (default $C = 2$) or the maximum round count $R_{\max}$ is reached.
5) **Output.** The accepted feature set $\mathcal{F}^*$, along with executable code and provenance cards for each feature.

B. Hierarchical Knowledge Injection

A key innovation of KnowFeat is the systematic integration of external domain knowledge into the feature generation process. We organize knowledge into five hierarchical levels of increasing specificity:

Level 0: Schema Only. The agent receives only the dataset schema—column names, data types, and basic statistics (mean, cardinality, missing rate). No domain knowledge is provided. This serves as the ablation baseline, testing what the LLM can generate from metadata alone.

Level 1: Risk Indicators. Structured risk indicators derived from regulatory guidelines and industry standards. Each indicator specifies a risk dimension, risk class, relevant data field, and directionality. For the AML domain, we compile 38 indicators (IND_01 through IND_38) covering transaction amount thresholds, frequency patterns, temporal anomalies, and network centrality measures.

Level 2: Detection Rules. Operational detection rules that encode specific logic patterns. Each rule includes a rule ID, natural language description, logical expression, confidence level, pattern category, and a Python implementation hint. We compile 15 rules (RULE_001 through RULE_015) from compliance manuals and internal documentation.

Level 3: Expert Opinions. Unstructured insights from domain expert interviews. These capture tacit knowledge—heuristics, edge cases, and domain intuitions—that are difficult to formalize as rules but valuable for feature design. Expert opinions are encoded as natural language paragraphs (EXP_01 through EXP_N).

Level 4: Court Document Evidence. Risk factors extracted from real legal proceedings using a key-clue extraction pipeline. For the AML domain, we process 309 court judgment documents to extract evidence-grounded risk factors (RF_S2_FFF_NNN format). This represents the highest level of domain specificity, grounding feature generation in real-world case evidence.

The hierarchy is *ablatable*: by setting a knowledge filter parameter, we can include any subset of levels, enabling systematic study of each level's contribution. In the "full" configuration (default), all levels are included. The knowledge base is formatted as structured text and injected into the agent's system prompt, with each knowledge item assigned a unique identifier for provenance tracking.

C. ReAct Agent with Specialized Tools

KnowFeat employs an LLM agent following the ReAct paradigm [14], interleaving reasoning (chain-of-thought) with actions (tool calls). The agent is equipped with three specialized tools:

Tool 1: `explore_data`. Executes arbitrary Python code on the dataset to inspect distributions, compute correlations, identify patterns, and understand the data before proposing features. This tool runs in a sandboxed environment and returns printed output or return values to the agent.

Tool 2: `generate_feature`. Registers a candidate feature with mandatory provenance fields:

- `feature_name`: A descriptive identifier.
- `code`: Executable Python code that computes the feature from raw data.
- `explanation`: A natural language description of what the feature captures.
- `reasoning`: A structured chain linking knowledge assets to the feature design, formatted as "Step N: Because [knowledge ID] indicates [insight], we compute [transformation]..."
- `knowledge_ids`: A list of specific knowledge asset identifiers (e.g., RULE_001, IND_05, EXP_03) that inspired the feature.
- `computation_steps`: Step-by-step description from raw fields to the final feature value.
- `data_fields_used`: The raw columns consumed by the feature.

Tool 3: `execute_sandbox`. Triggers the L1→L2→L3 verification pipeline on a registered feature and returns the result (accepted/rejected with detailed reasons). The agent must call `generate_feature` followed immediately by `execute_sandbox` for each candidate—batch registration is prohibited to ensure tight feedback loops.

The agent's prompt instructs it to: (1) reason about which knowledge assets are relevant to the current dimension and budget, (2) explore the data to validate assumptions, (3) propose a feature grounded in specific knowledge items, and (4) submit it for verification and react to the feedback.

D. Three-Stage Verification Pipeline

Each candidate feature passes through three progressively stricter verification stages:

1) L1: Execution Verification: The generated Python code is executed in an isolated sandbox with a 30-second timeout. L1 checks for:

- Syntax and runtime errors.
- Output shape consistency (one value per sample).
- Presence of required identifier columns.
- Data type validity (numeric output).

If L1 fails, the agent receives the error message and is allowed up to 2 automatic repair attempts before the feature is permanently rejected.

2) L2: Statistical Quality: The feature's statistical properties are evaluated against automatically calibrated thresholds. To establish these thresholds, we generate $N_{\text{noise}} = 100$ random features (drawn from standard distributions), compute their quality metrics, and set the acceptance threshold at the P_{noise}-th percentile (default $P_{\text{noise}} = 95$). This ensures that accepted features are statistically better than random noise. Specifically, L2 checks:

- **Information Value (IV)**: Must exceed the noise-calibrated threshold.
- **KS Statistic**: Must exceed the noise-calibrated threshold.
- **Redundancy**: Pearson correlation with all existing features must be below 0.7.
- **Missing Rate**: Must be below 30%.

3) L3: Model Effectiveness: The feature is evaluated for its actual contribution to predictive performance. We train an XGBoost model on the baseline features augmented with the candidate feature, using stratified 80/20 train-test splits across 5 random seeds $\{42, 123, 456, 789, 2024\}$. The acceptance criterion depends on the task:

- **Classification (public datasets)**: $\Delta\text{AUC} > 0$, i.e., the augmented model must achieve strictly higher AUC than the baseline.
- **AML detection**: $\Delta\text{Recall} > 0$ *and* $\text{Precision} \geq 90\%$, ensuring that the feature improves detection without sacrificing precision.

The three-stage design is intentionally progressive: L1 is cheap (milliseconds), L2 is moderate (seconds), and L3 is expensive (minutes per feature). By filtering early, we avoid wasting computational resources on clearly defective candidates.

E. Intelligent Search Strategy

To systematically explore the feature space, KnowFeat employs a multi-component search strategy.

1) Dimension-Based Budgeting: We decompose the feature space into $D = 4$ semantic dimensions relevant to the target domain:

1) **Amount & Frequency**: Features related to monetary values and transaction counts.
2) **Time & Geography**: Features capturing temporal patterns and spatial distributions.
3) **Behavior Sequence**: Features encoding sequential patterns in user actions.
4) **Entity Association**: Features derived from relationships between entities (accounts, counterparties).

Each dimension receives an initial budget of B_0 candidates per round (default $B_0 = 3$). After each round, SHAP (SHapley Additive exPlanations) analysis is performed on the current model to estimate each dimension's contribution to predictions.

2) SHAP-Guided Budget Reallocation: Let s_j denote the mean absolute SHAP value for dimension j's accepted features. We reallocate the budget using an *inverse-proportion* strategy:

$$B_j^{(r+1)} = B_{\min} + (B_{\text{total}} - D \cdot B_{\min}) \cdot \frac{1/s_j}{\sum_{j'=1}^{D} 1/s_{j'}} \quad (1)$$

where $B_{\min}$ is a minimum per-dimension budget (default 1) and B_{total} is the total budget for the round. This ensures that underrepresented dimensions—those contributing least to the current model—receive more exploration budget in subsequent rounds, promoting balanced feature coverage.

3) Failure Memory: A failure memory $\mathcal{M}_{\text{fail}}$ records all rejected features along with their rejection reasons. Before each generation step, the K most recent failures for the current dimension (default $K = 10$) are injected into the agent's prompt, preventing the regeneration of previously rejected ideas and guiding the agent toward novel directions.

TABLE II
DATASET STATISTICS.

Dataset	Samples	Feat.	Pos%	Domain
credit_g	1,000	20	70.0	Finance
diabetes	768	8	34.9	Healthcare
blood_transfusion	748	4	23.8	Healthcare
heart	918	11	55.3	Healthcare
bank_marketing	45,211	16	11.7	Marketing
breast_w	683	9	35.0	Healthcare
adult	48,842	14	23.9	Census
SimECNY (AML)	200,000	93	2.24	Finance

4) Forward Selection: Individual features may fail L3 verification (i.e., $\Delta\text{Recall} = 0$ or $\Delta\text{AUC} \leq 0$) yet contribute positively when combined with other candidates. To capture such synergistic effects, we apply greedy forward selection on the pool of L2-passed but L3-failed candidates:

1) Sort candidates by their L2 quality scores.
2) Greedily add each candidate to the current accepted feature set.
3) If the augmented set improves the target metric, accept the candidate; otherwise, discard it.

This mechanism is particularly important for high-dimensional datasets where individual features have marginal effects but feature combinations capture meaningful interactions.

5) Cross-Dimension Propagation: High-scoring features accepted in one dimension may inspire analogous features in other dimensions. After each round, the top-K_{cross} features (default $K_{\text{cross}} = 5$) by SHAP importance are propagated to other dimensions as "inspiration prompts," encouraging the agent to adapt successful patterns across different semantic contexts.

IV. EXPERIMENTS

We evaluate KnowFeat on both public tabular benchmarks and a real-world anti-money laundering (AML) dataset. Our experiments aim to answer four research questions:

- **RQ1**: Does KnowFeat improve predictive performance on public datasets?
- **RQ2**: How does KnowFeat perform on a real-world AML task with severe class imbalance?
- **RQ3**: What is the contribution of domain knowledge? (Ablation study)
- **RQ4**: How does KnowFeat balance feature quantity vs. quality?

A. Experimental Setup

1) Datasets: We use eight datasets summarized in Table II: seven public benchmarks from OpenML and one real-world AML dataset.

The seven public datasets are standard tabular benchmarks covering diverse domains. SimECNY is a real-world digital currency transaction dataset containing 200,000 transactions with 93 baseline features and 4,484 positive (suspicious)

samples, representing a realistic AML scenario with severe class imbalance (2.24% positive rate).

2) Baselines: We compare against six baselines:

- **No-FE**: XGBoost on original features.
- **OpenFE** [2]: Transformation-based AutoFE with LightGBM-based selection.
- **AutoFeat** [3]: Polynomial feature generation with L1 selection.
- **CAAFE** [4]: LLM-based iterative feature generation with logistic regression feedback.
- **FeatLLM** [5]: LLM-based rule extraction and binary feature generation.
- **LLM-FE** [8]: LLM-based evolutionary feature optimization that iteratively generates and refines features using structured prompts.

3) Evaluation Protocol: All methods are evaluated using XGBoost (n_estimators=200, max_depth=6, learning_rate=0.1) with stratified 80/20 train-test splits across 5 random seeds. For public datasets, the primary metric is AUC. For SimECNY, we report Recall, Precision, F1, and AUC with scale_pos_weight=auto ($\approx$43.6) to handle class imbalance.

B. RQ1: Public Dataset Performance

Table III presents the AUC comparison on seven public datasets. KnowFeat achieves the best performance on 4 out of 7 datasets (credit_g, blood_transfusion, breast_w, and adult) and ranks second on 2 datasets (diabetes and heart).

KnowFeat achieves the best average rank (1.4), substantially outperforming the second-best OpenFE (2.1). The largest improvements occur on datasets where domain knowledge provides strong semantic guidance: credit_g (+0.018 over No-FE) and blood_transfusion (+0.018 over No-FE).

OpenFE achieves the best results on 3 datasets (diabetes, heart, bank_marketing) through exhaustive operator-operand search. However, OpenFE generates 34–50 features per dataset, while KnowFeat achieves competitive or superior performance with 12–46 features (see Section IV-E).

Among LLM-based methods, LLM-FE generally outperforms CAAFE and FeatLLM but remains below both OpenFE and KnowFeat. On bank_marketing, LLM-FE even underperforms the No-FE baseline (.9317 vs. .9330), and on blood_transfusion it ranks last (.6649). This suggests that evolutionary prompt optimization alone, without external domain knowledge and rigorous multi-stage verification, is insufficient.

C. RQ2: AML Dataset Performance

Table IV presents results on the SimECNY AML dataset with scale_pos_weight=auto. KnowFeat adds 18 knowledge-guided features to the 93-feature baseline, achieving the best results across all four metrics.

KnowFeat's 18 domain-knowledge features yield substantial improvements: Recall improves by +2.1pp over No-FE (0.9871 vs. 0.9663), F1 by +1.5pp (0.9863 vs. 0.9713), and AUC reaches 0.9999. LLM-FE achieves the second-best F1 (0.9728) with only 1 additional feature, but KnowFeat's margin over LLM-FE remains large (+1.4pp F1, +1.7pp Recall). FeatLLM adds 1,031 features yet achieves the *worst* performance across all metrics.

To verify that KnowFeat's gains stem from intrinsic feature quality rather than interaction with class balancing, we re-evaluate all methods with scale_pos_weight=1 (no class balancing). KnowFeat remains the clear winner (Recall 0.9683, F1 0.9812), and is the *only* method that outperforms the No-FE baseline. KnowFeat suffers the smallest Recall drop when switching from spw=auto to spw=1 ($\Delta = -1.9$pp), while FeatLLM drops the most ($\Delta = -3.3$pp). This confirms that KnowFeat's 18 features encode genuine discriminative signal that does not depend on class reweighting.

D. RQ3: Ablation Study

To quantify the contribution of domain knowledge, we compare the *schema_only* variant (Level 0: only dataset metadata) against the *full* variant (all knowledge levels). Results on public datasets are shown in Table V.

Knowledge drives feature generation. On the diabetes dataset, the schema-only variant generates *zero* features that pass L3 verification. With full knowledge, KnowFeat generates 24 accepted features and improves AUC by +0.016 over No-FE.

Knowledge improves feature quality. On five of the six remaining datasets, the full variant outperforms schema_only, with gains ranging from +0.001 (adult) to +0.012 (blood_transfusion). On blood_transfusion, both variants generate 12 features, indicating that the improvement comes from feature *quality*, not quantity.

SimECNY ablation. On the AML dataset, domain knowledge is critical: schema_only generates only 3 features (+0.3pp Recall over No-FE), while the full knowledge hierarchy generates 18 features (+2.1pp Recall). The gap between schema_only and full (+1.8pp Recall, +1.4pp F1) confirms that the AML-specific knowledge assets (detection rules, behavioral patterns, expert opinions) are the primary driver of KnowFeat's advantage.

E. RQ4: Feature Quantity vs. Quality

Table VI compares the number of features generated by each method.

FeatLLM generates by far the most features (98–1,124) yet achieves the worst or near-worst performance on most datasets. AutoFeat generates 165 features on adult yet ranks last (AUC 0.9254). LLM-FE generates a conservative number of features (9–21) but cannot consistently outperform the No-FE baseline.

KnowFeat strikes an effective balance: it generates a moderate number of features (12–46 on public datasets, +18 on SimECNY) that have all passed the L1$\rightarrow$L2$\rightarrow$L3 pipeline. On SimECNY, the contrast is particularly stark: KnowFeat's 18 verified features improve F1 by +1.5pp, while FeatLLM's 1,031 unverified features *decrease* F1 by -0.4pp.

F. Provenance and Interpretability

Every KnowFeat-generated feature includes a provenance card recording: (1) the knowledge assets that inspired it; (2) a

TABLE III
AUC COMPARISON ON PUBLIC DATASETS (MEAN±STD OVER 5 SEEDS). **BOLD**: BEST; UNDERLINE: SECOND BEST. T/O = TIMEOUT.

Dataset	No-FE	OpenFE	AutoFeat	CAAFE	FeatLLM	LLM-FE	KnowFeat
credit_g	.7670±.021	.7833±.031	.7540±.017	.7596±.023	.7604±.022	.7615±.026	**.7854±.014**
diabetes	.8027±.033	**.8252±.034**	.8089±.027	.8032±.030	.8096±.032	.8148±.033	.8190±.032
blood_transfusion	.6824±.043	.6654±.040	.6824±.043	.6779±.043	.6805±.044	.6649±.045	**.7007±.036**
heart	.9320±.007	**.9417±.011**	.9340±.003	.9302±.006	.9297±.004	.9330±.005	.9379±.006
bank_marketing	.9330±.002	**.9374±.002**	T/O	.9334±.002	.9337±.002	.9317±.002	.9344±.002
breast_w	.9926±.005	.9944±.005	.9936±.004	.9924±.006	.9924±.005	.9939±.001	**.9957±.003**
adult	.9288±.004	.9300±.003	.9254±.004	.9286±.003	.9289±.003	.9275±.003	**.9301±.003**
Avg. Rank	4.5	2.1	5.1	5.5	4.6	4.7	**1.4**

TABLE IV
SIMECNY AML RESULTS (SPW=AUTO, 5 SEEDS).

Method	n_f	Recall	Prec.	F1	AUC
No-FE	93	.9663±.010	.9764±.004	.9713±.005	.9996
OpenFE	123	.9645±.008	.9752±.004	.9698±.003	.9996
CAAFE	94	.9681±.008	.9725±.006	.9703±.005	.9996
FeatLLM	1124	.9625±.010	.9728±.005	.9676±.006	.9995
LLM-FE	94	.9699±.007	.9758±.004	.9728±.005	.9996
Ours	**111**	**.9871**±.006	**.9855**±.001	**.9863**±.003	**.9999**

TABLE V
ABLATION: SCHEMA_ONLY VS. FULL KNOWFEAT (AUC).

Dataset	schema_only	full	Δ
credit_g	.7750±.012	**.7854**±.014	+.010
blood_trans.	.6891±.048	**.7007**±.036	+.012
heart	.9304±.005	**.9379**±.006	+.008
bank_mkt.	.9314±.002	**.9344**±.002	+.003
adult	.9292±.003	**.9301**±.003	+.001
breast_w	**.9964**±.002	.9957±.003	−.001
diabetes	— (0 feat.)	.8190±.032	—

reasoning chain explaining *why* the feature is relevant; and (3) the computation steps from raw fields to the final value. This traceability is essential for deployment in regulated industries. We illustrate this capability through detailed case studies in Section V.

V. CASE STUDY: KNOWLEDGE-GROUNDED FEATURES

To illustrate how KnowFeat leverages domain knowledge to generate interpretable, high-quality features, we present three representative case studies drawn from the SimECNY AML dataset. Each example highlights a different aspect of knowledge-guided feature engineering: multi-source knowledge fusion, domain-specific behavioral patterns, and cross-dimensional interaction.

A. Case 1: Fan-In/Fan-Out Ratio (Network Topology)

Feature: `fan_in_fan_out_ratio`, which measures the asymmetry between incoming and outgoing transaction patterns, weighted by the proportion of abnormal transaction modes.

TABLE VI
NUMBER OF FEATURES (INCL. ORIGINAL).

Dataset	No-FE	Open. FE	Auto. Feat	CAA. FE	Feat. LLM	LLM- FE	Ours
credit_g	20	50	23	24	280	21	46
diabetes	8	38	13	9	152	15	24
blood_trans.	4	34	4	7	120	9	12
heart	11	41	21	14	98	19	31
bank_mkt.	16	46	T/O	21	132	20	36
breast_w	9	39	20	10	107	15	16
adult	14	44	165	19	170	21	34
SimECNY	93	123	—	94	1124	94	111

Knowledge assets used:

- *RULE_006* (Detection Rule): "Scatter-in, gather-out," a classic layering pattern where funds are collected from many sources then concentrated for withdrawal.
- *RULE_013* (Detection Rule): Pyramid-shaped fund networks where intermediate wallets exhibit strong fan-in/fan-out asymmetry.
- *PAT_A03* (Behavioral Pattern): Online money mule wallets receive from many senders (high fan-in) then forward to a few consolidation accounts.
- *IND_15* (Risk Indicator): Abnormal transaction modes (fan_in, many_to_many, one_to_many) are individually suspicious.

Reasoning chain:

1) From *RULE_006* and *RULE_013*: compute the ratio of fan-in to fan-out mode proportions per wallet.
2) From *PAT_A03*: money mule wallets have fan-in $\gg$ fan-out; this asymmetry ratio is the primary signal.
3) From *IND_15*: weight by total abnormal mode proportion, since wallets with more abnormal modes are higher risk.
4) Combine: ratio $\times$ (1 + abnormal_mode_ratio).

Generated code (simplified):

```
fi = mode_counts['fan_in'] / n
fo = mode_counts['fan_out'] / n
ratio = fi / (fo + eps)
abnormal = fi + m2m_ratio + o2m_ratio
result = ratio * (1 + abnormal)
```

Why this matters. This feature fuses knowledge from four assets spanning three types (rules, patterns, indicators) to capture network topology. No individual column contains "fan-in/fan-out asymmetry"—it requires understanding that the *combination* of directional imbalance and abnormal mode prevalence characterizes money mule behavior. A syntactic method might discover pairwise mode interactions but would not produce a topology-aware asymmetry score grounded in the "scatter-in, gather-out" layering pattern.

B. Case 2: Probe-Then-Transfer Ratio

Feature: `small_test_round_amount_ratio`, which detects the "probe-then-bulk-transfer" pattern by combining small-amount test ratio with round-number transfer ratio.

Knowledge assets used:

- *RULE_003* (Detection Rule): Rapid fund movement—small probe transactions ($\leq$5 CNY) precede large-value transfers.
- *PAT_A02/A03* (Behavioral Pattern): Money mule operations begin with "small test, then bulk operation."
- *IND_01/02* (Risk Indicator): Round-number amounts (multiples of 100) and repetitive amount patterns indicate structured transactions.

Reasoning chain:

1) From *RULE_003* and *PAT_A02/A03*: compute the proportion of transactions with amount $\leq$5 CNY (probe ratio).
2) From *IND_01/02*: compute the proportion of transactions with amounts divisible by 100 (round-number ratio).
3) From domain expertise (*EXP_01*): "probe + round-number bulk transfer" co-occurrence is far more discriminative than either signal alone.
4) Combine: small_ratio $\times$ (1 + round_100_ratio).

Generated code (simplified):

```
amt = abs(amount)
small_ratio = (amt <= 5).sum() / n
round_ratio = (amt % 100 == 0).sum() / n
result = small_ratio * (1 + round_ratio)
```

Why this matters. The "probe-then-transfer" pattern is well-documented in AML regulations but impossible to discover through syntactic transformation search. It requires understanding that sub-5-CNY transactions serve a fundamentally different purpose (channel testing) than regular small purchases. The specific threshold (5 CNY) and the interaction with round-number amounts come directly from *RULE_003* and compliance expertise—transformations that no data-driven method would enumerate.

C. Case 3: Cross-Dimensional Risk Interaction

Feature: `high_risk_region_off_hour_ratio`, a multiplicative interaction between geographic risk and temporal anomaly.

Knowledge assets used:

- *RULE_005/014* (Detection Rules): Cross-regional fund flows involving border areas (region codes 540000/650000) are high-risk for underground banking.
- *IND_08/09* (Risk Indicators): Transactions concentrated in off-hours (18:00–8:00) indicate abnormal activity.
- *EXP_05* (Expert Opinion): "Border region + nighttime trading is the hallmark of cross-border money laundering."

Reasoning chain:

1) From *RULE_005/014*: flag wallets registered in high-risk border regions.
2) From *IND_08/09*: compute off-hour transaction ratio.
3) From *EXP_05*: neither signal alone is sufficient—border wallets are not inherently suspicious, nor are off-hour transactions. The *combination* is the strong signal.
4) Combine via multiplication: is_high_risk $\times$ off_hour_ratio. Non-border wallets receive 0 regardless of timing.

Generated code (simplified):

```
is_hr = region.isin([540000, 650000])
off = (hour >= 18) | (hour < 8)
off_ratio = off.sum() / n
result = is_hr * off_ratio
```

Why this matters. This feature demonstrates *cross-dimensional interaction*: it combines geographic context (from account metadata) with temporal patterns (from transaction logs), guided by expert knowledge that explicitly identifies the combination as a "hallmark" signal. Syntactic methods cannot cross data sources or reason about why border + nighttime is meaningful. The multiplicative form ensures that neither dimension alone triggers a high score—only the conjunction, exactly as the expert knowledge prescribes.

TABLE VII
SUMMARY OF CASE STUDY FEATURES.

Feature	#KA	Key Insight
`fan_in_fan_out`	4	Network topology fusion
`small_test_round`	5	Behavioral sequence pattern
`region_off_hour`	4	Cross-dim. interaction

KA = Knowledge Assets in provenance chain.

D. Summary of Case Study Findings

Table VII summarizes the three case studies.

These examples demonstrate three capabilities that distinguish KnowFeat: (1) *multi-source fusion*—combining rules, behavioral patterns, and risk indicators into topology-aware features; (2) *domain-specific patterns*—encoding regulatory knowledge (probe thresholds, structuring indicators) that syntactic search cannot discover; and (3) *cross-dimensional interaction*—combining signals from different data sources guided by expert knowledge about their conjunction. Every feature includes a complete provenance chain, enabling compliance officers to audit the feature's rationale.

VI. Conclusion

We presented KnowFeat, a knowledge-guided feature engineering framework that combines hierarchical domain knowledge injection, multi-stage verification (L1–L3), and intelligent search strategies within an LLM agent. On seven public datasets, KnowFeat achieves the best AUC on 4 out of 7 and the best average rank (1.4) among seven methods, while generating 3–10$\times$ fewer features than FeatLLM. On a real-world AML dataset with 200K transactions, 18 knowledge-guided features improve Recall by +2.1pp and F1 by +1.5pp over a strong 93-feature baseline—gains that are robust across different class balancing settings. The ablation study confirms that domain knowledge is essential, and every generated feature is traceable to specific knowledge assets.

Limitations and Future Work. First, our current knowledge hierarchy is manually organized; automating knowledge extraction from heterogeneous sources (regulations, case law, expert interviews) is a promising direction. Second, KnowFeat currently uses a single LLM; a multi-agent architecture with specialized agents for knowledge retrieval, feature proposal, and verification could improve efficiency. Third, extending KnowFeat to additional high-stakes domains such as cybersecurity and insurance fraud detection would validate the generality of the approach. Finally, investigating adaptive L3 thresholds to balance feature quantity and quality across different dataset characteristics is an important direction.

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